

A Multi-stage Bioinformatics and Structural Prioritization Pipeline for Membrane Protein Anchor Discovery in Idiopathic Pulmonary Fibrosis

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Idiopathic pulmonary fibrosis (IPF) is a progressive fibrotic lung disease in which disease-associated fibroblasts and myofibroblasts are central effector cells. Targeted delivery to these populations requires membrane-accessible anchors that combine disease upregulation, target-cell enrichment, extracellular accessibility, ligand or binder feasibility and manageable signaling risk. Here, we propose a bioinformatics and structure-informed prioritization workflow for IPF membrane-anchor discovery. Starting from 52,971 human protein entries, a broad membrane-associated library of 7,717 entries was constructed. GSE32537 analysis identified 1,089 IPF-upregulated genes; intersection with the membrane library and HPA plasma-membrane annotation yielded 131 candidates. GSE135893 single-cell evidence refined six priority candidates to four, deprioritizing NRP2 and APLNR for insufficient fibroblast enrichment. Ligand-receptor feasibility and pathway-risk assessment prioritized PLXDC1 and ROBO1, retaining three model hypotheses: PEDF-PLXDC1, SLIT2-ROBO1 and SLIT3-ROBO1. This study provides a reproducible computational framework for early-stage membrane-anchor prioritization and generates testable hypotheses for future IPF-targeted delivery studies. All analysis scripts are publicly available at <https://github.com/yuanqianmingbeta-gif/ipf-membrane-pipeline>.

Additional Key Words and Phrases: idiopathic pulmonary fibrosis, membrane protein anchor, fibroblast, structural bioinformatics

1 INTRODUCTION

Idiopathic pulmonary fibrosis is a chronic, progressive fibrosing interstitial pneumonia associated with excessive extracellular matrix deposition, architectural remodeling of the lung and poor prognosis. Although approved anti-fibrotic drugs can slow disease progression, curative options remain limited and lung transplantation remains the only definitive treatment for end-stage disease [1, 2].

Fibroblasts and myofibroblasts are central contributors to IPF progression because they produce extracellular matrix, promote tissue contraction and maintain a pro-fibrotic microenvironment. Selective delivery of therapeutic payloads to these populations therefore requires cell-surface membrane anchors that are accessible from the extracellular space and sufficiently enriched in disease-relevant stromal cells.

Conventional target-discovery strategies prioritize differential expression or pathway relevance, but delivery-anchor discovery requires additional criteria: membrane topology, extracellular exposure, target-cell localization, ligand feasibility and acceptable activation risk. Bulk transcriptomics identifies disease-associated genes, but single-cell data verify that expression originates from fibroblast/myofibroblast populations [3–5].

We integrated curated protein annotation, bulk IPF upregulation, HPA expression evidence, manual anchor curation, single-cell refinement and structure-guided ligand-receptor analysis to produce a reproducible workflow generating structurally testable anchor hypotheses for IPF-targeted delivery.

2 MATERIALS AND METHODS

The overall multi-stage workflow for membrane-anchor discovery, including computational screening, biological refinement, ligand–receptor assessment and structural analysis, is summarized in Figure 1.

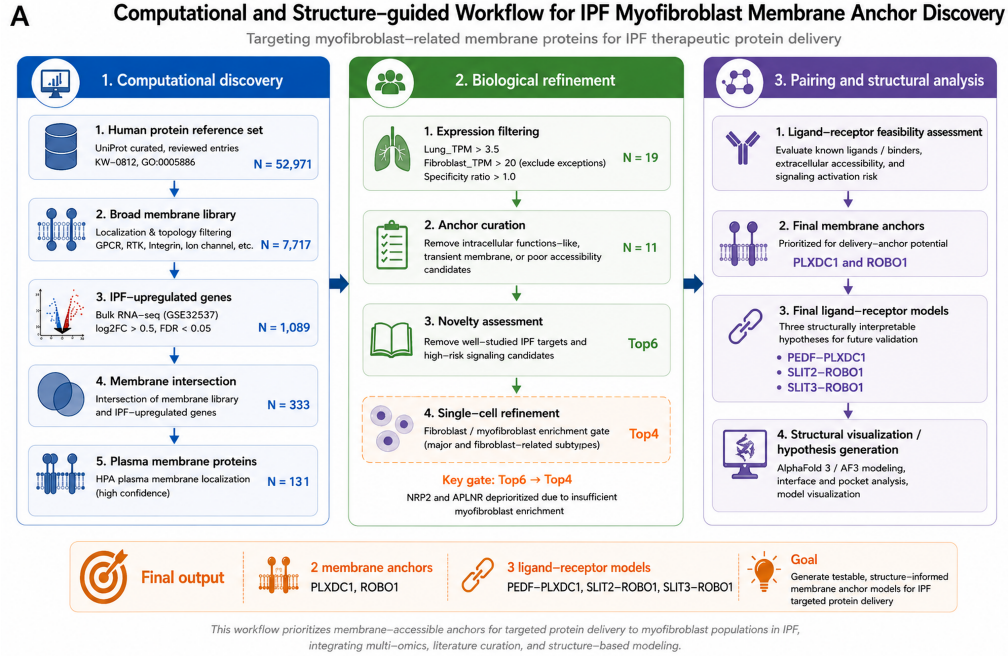


Fig. 1. Computational and structure-guided workflow for IPF membrane-anchor discovery. Each box annotates the data source, filtering criteria, and candidate count. Structural tools: AlphaFold3, Foldseek, PISA, ConSurf, fpocket, ChimeraX.

2.1 Human Protein Library and Membrane Localization Refinement

A master human protein reference set was assembled from curated protein annotation tables containing gene symbol, protein name, UniProt accession, subcellular localization, transmembrane topology, sequence and functional information. UniProtKB was used as the major annotation source because it provides curated protein sequence and functional annotation [6]. The initial membrane-associated library was intentionally broad: proteins were retained using human and membrane-related annotation without restricting the search to plasma-membrane localization. This broad membrane-associated library contained 7,717 protein entries.

Because delivery-anchor discovery requires extracellular accessibility, plasma-membrane localization was treated as a later refinement step rather than as the initial library-construction criterion. After intersection with IPF-upregulated genes, proteins were further refined using HPA subcellular localization annotation; only proteins annotated with “plasma membrane” in HPA subcellular location were retained for the upregulated plasma-membrane candidate set [7]. This broad-to-specific strategy was used to avoid losing potentially relevant membrane-associated proteins during initial database construction, while ensuring extracellular accessibility during candidate prioritization.

2.2 GSE32537 Differential Expression Analysis

GSE32537 (GPL6244, Affymetrix Human Gene 1.0 ST Array) contains 119 IPF/UIP and 50 non-diseased control lung tissue samples with clinical metadata (FVC, DLCO, smoking status) from the Lung Tissue Research Consortium [3]. GSE32537 was analyzed using a Python-based differential-expression workflow. The GEO series matrix file was downloaded programmatically, and sample metadata were parsed from sample title, source and characteristic fields. IPF/UIP lung

tissue samples were assigned to the case group, and non-diseased lung samples were assigned to the control group using keyword-based sample classification. If the expression matrix appeared to be non-log-transformed, values were transformed as $\log_2(x + 1)$.

For each probe, mean expression was calculated in the case and control groups, and log2 fold change was computed as the difference between the two group means. Statistical significance was assessed using Welch's two-sample t-test, and P values were adjusted using the Benjamini-Hochberg procedure. Probe identifiers were mapped to gene symbols using the corresponding GEO platform annotation. Probe-level differential-expression results were deduplicated at the gene-symbol level by retaining the record with the lowest adjusted P value and, secondarily, the highest log2 fold change. Genes with $\log_2\text{FC} > 0.5$ and adjusted P value < 0.05 were considered IPF-upregulated, yielding 1,089 deduplicated upregulated genes.

The 1,089 IPF-upregulated genes were intersected with the 7,717-entry broad membrane-associated library, yielding 333 upregulated membrane-associated genes. These 333 genes were then refined using the HPA plasma-membrane localization criterion described above, producing 131 upregulated plasma-membrane candidate proteins.

2.3 HPA Expression Filtering and Candidate Prioritization

HPA-derived expression evidence was used to prioritize candidates with lung expression and fibroblast-associated expression. Lung expression was evaluated using lung TPM, and fibroblast-associated enrichment was assessed using fibroblast nTPM and a specificity ratio defined as fibroblast nTPM divided by the mean nTPM of other cell types. Three thresholds were selected based on HPA tissue-specificity classifications [7]: lung TPM > 3.5 ensures detectable lung expression above background; fibroblast nTPM > 20 corresponds to the HPA "medium" single-cell expression category, ensuring biologically meaningful fibroblast expression; and specificity ratio > 1.0 requires that fibroblast expression exceeds the mean of other cell types. These three thresholds were used as the main expression-support criteria. Fibroblast nTPM was treated as a soft threshold with exceptions for ligand-feasible receptors, which were retained during exploratory comparison before being tested by the single-cell refinement gate.

2.4 Manual Curation, Single-cell Gate and Model Prioritization

The 19 expression-filtered proteins were manually reviewed for practical anchor feasibility. Candidates were removed if they showed mainly intracellular functional orientation, unstable or transient membrane localization, insufficient extracellular exposure or poor ligand/binder accessibility. Literature review then evaluated fibrosis relevance, ligand or interface evidence, IPF novelty and pathway activation risk. This generated a top-six candidate set: PLXDC1, NRP2, APLNR, HEPH, ROBO1 and BOC.

GSE135893 provides scRNA-seq data from 20 PF and 10 control lungs (10X Genomics; 114,396 cells, 31 cell types) [4]. GSE135893 single-cell RNA-seq evidence was used as the decisive top-six to top-four refinement step. Candidates were scored for fibroblast/myofibroblast enrichment using a 0/3/5 scale: 5 for strong target-cell enrichment, 3 for partial enrichment and 0 for insufficient target-cell support or dominant non-target expression. A score of 0 triggered automatic deprioritization regardless of the total ranking score. Ligand-receptor feasibility was assessed by ligand availability, binding specificity, interface evidence, engineering feasibility, delivery relevance and activation-risk penalty. Structural interpretation used AlphaFold-predicted monomer structures [8] retrieved from the AlphaFold Protein Structure Database [9], AlphaFold3/complex modeling where available [10], Foldseek homology search [11], and ChimeraX visualization [12]. Interface residues were identified using the PISA web server [13]. Evolutionary conservation was estimated using ConSurf [14]. Ligand-binding pocket detection was performed using fpocket [15].

Table 1. Essential screening stages and retained candidates.

Stage	Main criterion	Output	Purpose
Human protein reference set	Human protein reference entries	52,971	Initial search space
Broad membrane-associated library	Human and membrane-related annotation; not restricted to plasma membrane	7,717	Capture membrane-associated proteins broadly
IPF-upregulated genes	GSE32537 Python analysis; log2FC > 0.5; adjusted P < 0.05	1,089	Identify disease-upregulated genes
Upregulated membrane-associated genes	Intersection of broad membrane library and IPF-upregulated genes	333	Restrict disease signal to membrane-associated genes
Upregulated plasma-membrane proteins	HPA subcellular localization contains “plasma membrane”	131	Retain extracellularly accessible anchor candidates
Expression-filtered candidates	lung TPM > 3.5; fibroblast nTPM > 20; specificity ratio > 1.0	19	Prioritize lung/fibroblast-associated proteins
Manual anchor curation	Remove poor extracellular accessibility, transient membrane candidates or intracellular functional orientation	11	Improve practical anchor feasibility
Novelty/risk assessment	Remove well-studied or high-risk candidates	Top6	Prioritize underexplored anchor candidates
Single-cell refinement	GSE135893 fibroblast/myofibroblast enrichment; score 0 excluded	Top4	Define cell-type-supported candidates
Pairing and structure assessment	Ligand evidence plus pathway-risk assessment	2 anchors / 3 models	Select model-supported structural hypotheses

2.5 Semi-quantitative Scoring and Risk Assessment

Candidate prioritization used a semi-quantitative ranking system with eight dimensions scored 0–2 (IPF upregulation, lung expression, fibroblast expression, bulk specificity, membrane-anchor feasibility, ECD accessibility, structural confidence, ligand/interface evidence), a single-cell enrichment score (0/3/5), and a risk penalty (0 to –3). The total score = $\Sigma(8 \text{ dimensions}) + \text{sc score} + \text{risk}$, with a maximum of 21. Each dimension was scored 0 (absent/insufficient), 1 (moderate/borderline), or 2 (strong/positive) based on predefined criteria: expression thresholds (lung TPM > 3.5, fibroblast nTPM > 20, specificity ratio > 1.0), structural confidence (ECD pLDDT ≥ 70 for score 2), membrane topology, and ligand evidence from known interactions or structural templates. Table 3 presents the per-dimension breakdown. The total numerical score was used only as a ranking aid, not as proof of biological validity. Because cell-type localization was central to the delivery-anchor objective, the GSE135893 single-cell gate was treated as decisive: any candidate assigned a single-cell score of 0 was deprioritized after single-cell refinement even if its total ranking score remained numerically moderate.

3 RESULTS

3.1 Candidate Space Construction and Top-six Selection

Figure 1 summarizes the three-phase workflow, and Table 1 lists the candidate counts at each filtering stage. The pipeline progressively narrowed the search space from 52,971 human protein entries (UniProt) to a broad membrane-associated library of 7,717 entries (GO and Human Protein Atlas). Python-based GSE32537 differential-expression analysis identified 1,089 IPF-upregulated genes. Intersecting the upregulated genes with the broad membrane-associated library yielded 333 upregulated membrane-associated genes. Because a delivery anchor must be accessible from the extracellular space, HPA subcellular localization was then used to retain candidates annotated with plasma-membrane localization, reducing the set to 131 upregulated plasma-membrane proteins.

HPA expression filtering further prioritized candidates with lung expression and fibroblast-associated specificity, producing 19 expression-supported candidates. Manual anchor-feasibility curation reduced this set to 11 proteins. Literature-based novelty and risk assessment generated six priority candidates: PLXDC1, NRP2, APLNR, HEPH, ROBO1

Table 2. Top-six candidate prioritization and single-cell refinement outcome.

Gene	Alias	scRNA status	sc score	Risk	Ranking score	Decision
PLXDC1	TEM7	Retained Top4	5	-2	18	Prioritized anchor hypothesis; PEDF-PLXDC1 model
NRP2	Neuropilin-2	Deprioritized after GSE135893	0	-3	13	Comparative candidate; insufficient target-cell specificity
APLNR	APJ	Deprioritized after GSE135893	0	-3	10	Comparative candidate; endothelial-skewed support
HEPH	Hephaestin	Retained Top4	5	-2	17	Top4 candidate; no final native ligand-receptor model
ROBO1	Roundabout guidance receptor 1	Retained Top4	5	-2	18	Prioritized anchor hypothesis; SLIT2/SLIT3-ROBO1 models
BOC	Brother of CDO	Retained Top4	5	-3	16	Top4 candidate; excluded from final native-ligand modeling due to Hedgehog-pathway risk

Note: Candidates with single-cell enrichment score = 0 were automatically deprioritized after GSE135893 refinement regardless of total numerical score. The ranking score was used as a prioritization aid rather than as experimental evidence of target validity.

Table 3. Ten-dimension scoring breakdown for the top-six candidates.

Gene	IPF	Lung	Fib	Spec	Anc	ECD	Str	Lig	sc	Risk	Total
PLXDC1	2	2	2	2	2	2	1	2	5	-2	18
NRP2	2	2	2	2	2	2	2	2	0	-3	13
APLNR	2	2	0	2	2	1	2	2	0	-3	10
HEPH	2	2	2	2	2	2	2	0	5	-2	17
ROBO1	2	2	2	1	2	2	2	2	5	-2	18
BOC	2	2	2	2	2	2	2	0	5	-3	16

Note: Eight dimensions scored 0–2 (0 = absent/insufficient, 1 = moderate/borderline, 2 = strong/positive); IPF = IPF upregulation, Lung = lung expression, Fib = fibroblast expression, Spec = bulk specificity, Anc = membrane anchor feasibility, ECD = ECD accessibility, Str = structural confidence, Lig = ligand/interface evidence; sc = single-cell enrichment (0/3/5); Risk = penalty (0 to -3). Total = $\sum(8 \text{ dimensions}) + \text{sc} + \text{risk}$. Scores below 2: PLXDC1 Str = 1 (ECD pLDDT 73.6, borderline); ROBO1 Spec = 1 (specificity ratio 1.18, near boundary); APLNR Fib = 0 (nTPM 1.6, below threshold); APLNR ECD = 1 (7TM GPCR, short loops); HEPH/BOC Lig = 0 (no native delivery-oriented ligand pair).

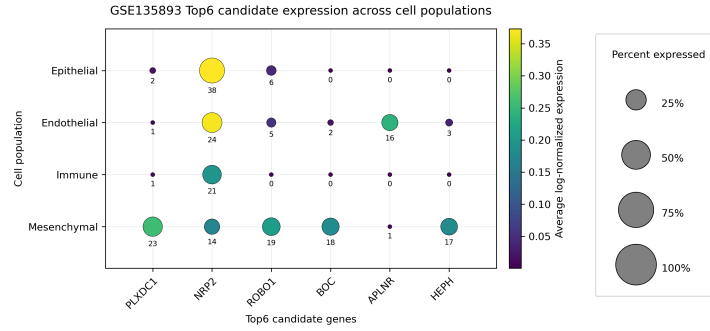
and BOC. Overall, the filtering funnel reduced the initial proteome by approximately 400-fold, from 52,971 entries to 131 plasma-membrane candidates, before biological refinement selected the final six anchors.

3.2 GSE135893 Refinement Identified the Top-four Candidates

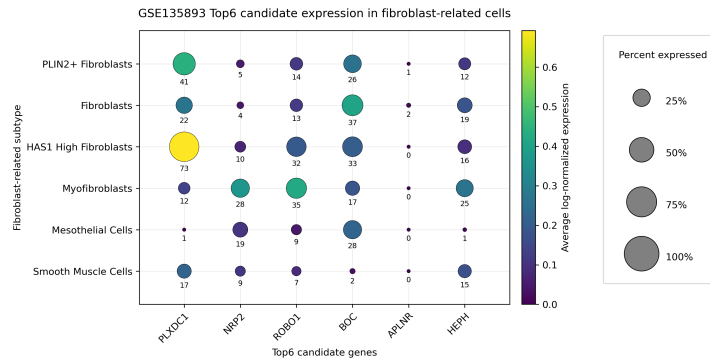
The top-six candidates were assessed in GSE135893 at both major cell-population and fibroblast-related subtype levels (Figure 2). PLXDC1, ROBO1, BOC and HEPH showed stronger mesenchymal/fibroblast-related support or specificity and were retained as the single-cell-supported top-four candidates. NRP2 showed stronger non-target expression, especially in epithelial/endothelial contexts, and APLNR showed insufficient target fibroblast/myofibroblast support. Both were therefore removed by the single-cell gate. At the subtype level (5,232 mesenchymal cells, six subtypes), PLXDC1 showed 41.5% detection in PLIN2+ fibroblasts vs. 22.8% population average, and BOC had the highest specificity ratio (59.3). NRP2 (0.63) and APLNR (0.22) fell below 1.0, confirming insufficient specificity.

3.3 Structural Feasibility of the Top-six Candidates

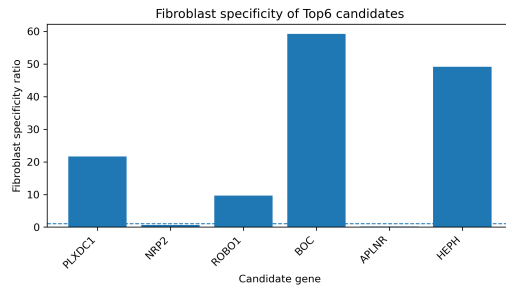
The top-six candidates differed substantially in topology and extracellular-domain accessibility (Table 4). ROBO1, NRP2, PLXDC1, BOC and HEPH were single-pass or signal-peptide-containing membrane proteins with large extracellular regions, whereas APLNR was a seven-transmembrane GPCR with extracellular accessibility limited to shorter extracellular loops. For ROBO1 and BOC, full-length average pLDDT was moderate, but the extracellular regions used for anchor interpretation had higher confidence. This supported domain-level interpretation while also indicating that full-length conformational arrangement should not be overinterpreted. Foldseek homology search corroborated the predicted structures: ROBO1 Ig1 matched PDB 2V9T (E-value = 2.6×10^{-18}) and APLNR matched 6KNM (E-value = 7.7×10^{-34}), providing experimental templates for interface interpretation.



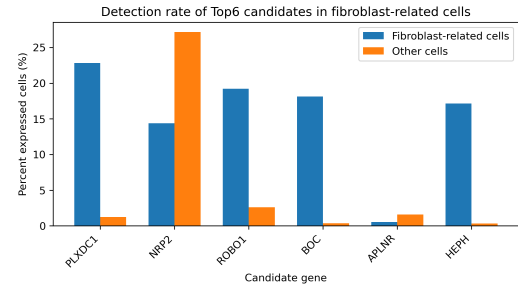
(a) Top-six candidate expression across major cell populations.



(b) Top-six candidate expression across fibroblast-related subtypes.



(c) Fibroblast specificity ratio of the top-six candidates.



(d) Detection-rate comparison between fibroblast-related and other cells.

Fig. 2. Single-cell refinement of candidate anchors using GSE135893.

3.4 Ligand-receptor Model Selection

After single-cell refinement and ligand-receptor feasibility assessment, PLXDC1 and ROBO1 were prioritized as model-supported anchor hypotheses. PLXDC1 was prioritized because it is an experimentally identified cell-surface receptor for PEDF, and PEDF binding depends on the extracellular domain of PLXDC1 [16]. PEDF has also been detected in IPF fibroblastic foci [17] and has demonstrated anti-fibrotic effects in a bleomycin-induced pulmonary fibrosis model

Table 4. Topology and extracellular-domain structural feasibility of the top-six candidates.

Candidate	UniProt	Length	Topology	ECD range	ECD pLDDT	Avg. pLDDT	Interpretation
APLNR	P35414	380	7TM GPCR	ECLs	80.43	81.69	Ligand-feasible GPCR, but extracellular access is limited to loops rather than a large ECD
ROBO1	Q9Y6N7	1651	SP+TM	26–897	85.21	60.00	Large ECD with interpretable domain-level confidence; full-length packing uncertain
NRP2	O60462	931	SP+TM	23–864	81.48	78.62	Large ECD and strong ligand biology, but single-cell specificity was insufficient
PLXDC1	Q8IUK5	500	SP+TM	19–426	73.59	72.19	Single-pass receptor with analyzable ECD and PEDF-binding rationale
BOC	Q9BWV1	1114	SP+TM	31–855	74.48	63.41	Large ECD, but native Hedgehog ligand context increases activation risk
HEPH	Q9BQS7	1158	SP+TM	24–1110	91.08	89.06	High-confidence extracellular catalytic region, but no clear native delivery-oriented ligand pair

Table 5. Summary of retained ligand-receptor structural model hypotheses.

Model	Input / version	ipTM / pTM	Area / ΔiG	H-bonds / salt bridges	Interpretation
PEDF-PLXDC1	Mature PEDF + PLXDC1 model V3.0	0.21 / 0.69	1011.7 / -15.5	4 / 0	Primary PLXDC1 delivery-oriented hypothesis; requires binding validation
SLIT2-ROBO1	2V9T-informed model V1.0	0.30 / 0.69	810.3 / 2.5	6 / 10	Structurally informed ROBO1 pairing based on known Slit-Robo interface precedent
SLIT3-ROBO1	SLIT3 D2-plus model V4.0	0.36 / 0.68	699.3 / -5.4	5 / 0	Comparative ROBO1 model with greater remodeling and activation uncertainty

Note: Interface area is reported in $\text{\AA}^2$ and ΔiG in kcal/mol. The ipTM values were modest; therefore, these complexes are interpreted as structure-guided hypotheses rather than validated binding modes. Experimental binding, internalization and signaling assays are required before delivery use.

[18]. ROBO1 was prioritized because the Slit-Robo interaction is structurally characterized: Slit2 D2 binds primarily to Robo1 Ig1 through a defined electrostatic and hydrophobic interface [19]. SLIT2 has been shown to inhibit fibrocyte differentiation and fibrosis [20], and N-terminal SLIT2 attenuates TGF- β -induced fibroblast activation [21].

The retained model hypotheses were PEDF-PLXDC1, SLIT2-ROBO1 and SLIT3-ROBO1 (Table 5, Figure 3). PEDF-PLXDC1 was interpreted as the primary delivery-oriented hypothesis; SLIT2-ROBO1 as a ROBO1 pairing with stronger precedent from the Slit-Robo structural literature; and SLIT3-ROBO1 as a higher-information comparison model with greater remodeling/activation uncertainty; the SLIT3-ROBO1 axis has been implicated in tissue remodeling [22]. HEPH remained in the top-four set but lacked a defined native extracellular ligand-receptor pair suitable for this modeling framework, and BOC was excluded from final native-ligand modeling because Hedgehog-associated ligand engagement [23] could directly activate a profibrotic developmental pathway [24].

3.5 Interface Hotspot Mapping

Integrating PISA interface residues, ConSurf conservation (150-sequence MSA per protein), and fpocket pocket detection identified 258 total hotspot residues across the three complexes, of which 17 met strict high-priority criteria (PISA interface + high BSA + ConSurf grade ≥ 7 + fpocket overlap): 10 in PEDF-PLXDC1, four in SLIT2-ROBO1, and three in SLIT3-ROBO1. The PLXDC1 cluster (residues 149–156) forms a contiguous extracellular patch; the ROBO1 Ig1 hotspots overlap the 2V9T-defined SLIT2-binding region.

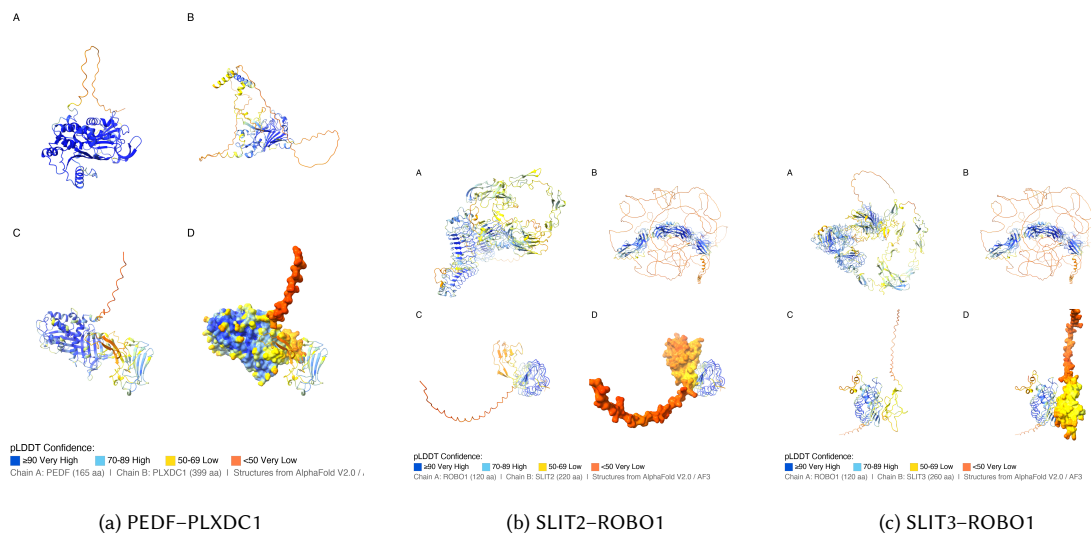


Fig. 3. Structural models of the three retained ligand–receptor complexes. Each panel shows a four-view composite (monomer A, monomer B, complex cartoon, complex surface) colored by per-residue pLDDT confidence: dark blue (≥ 90 , very high), light blue (70–89, high), yellow (50–69, low), orange (< 50 , very low). (a) PEDF–PLXDC1: Chain A (PEDF, 165 aa), Chain B (PLXDC1, 399 aa). (b) SLIT2–ROBO1: Chain A (SLIT2 D2, 220 aa), Chain B (ROBO1 Ig1–2, 120 aa). (c) SLIT3–ROBO1: Chain A (SLIT3 D2, 220 aa), Chain B (ROBO1 Ig1–2, 120 aa). Monomer structures were predicted using AlphaFold V2.0; complex models were generated using AlphaFold3.

4 DISCUSSION

This study proposes a membrane-anchor discovery workflow distinct from conventional target-discovery: a delivery anchor must be disease-relevant, cell-surface-accessible, target-cell-enriched and compatible with an extracellular recognition module.

4.1 Broad Membrane Discovery Followed by Plasma-membrane Refinement

The distinction between broad membrane-associated proteins and plasma-membrane-accessible candidates is critical. The 7,717-entry library was intentionally broad; after intersection with IPF-upregulated genes (333) and HPA plasma-membrane refinement, 131 candidates remained. This step is essential because intracellular organelle membrane proteins are not accessible to extracellular delivery.

4.2 Why Established IPF Targets Were Not Retained

Several established IPF targets were not retained. Most were absent from the GSE32537 upregulated list or lacked fibroblast specificity: CXCR4 (specificity ratio 0.12; monocyte nTPM 2,202 vs. fibroblast 19.8), ITGB6 (AT2-cell dominant; nTPM 105.6 vs. fibroblast 7.9), and IL6R (ratio 0.79). EDNRA passed all filters ($\log_2\text{FC} = 0.56$, ratio = 2.25) but was excluded as a mature clinical target: ambrisentan, a selective endothelin receptor A antagonist, was evaluated in a randomized IPF trial and did not demonstrate efficacy [25]. EDNRA was therefore retained only for comparison. These exclusions reflect delivery-anchor requirements—accessibility, enrichment, novelty and manageable risk—not biological irrelevance.

4.3 Value of the Single-cell Gate

The two-level refinement strategy is the strongest methodological feature: bulk GSE32537 analysis identified upregulated genes, while GSE135893 single-cell data verified fibroblast/myofibroblast enrichment. NRP2, despite roles in TGF- β 1-induced EMT [26], showed endothelial/epithelial off-target expression. APLNR, despite apelin's anti-fibrotic potential [27], lacked single-cell fibroblast support.

4.4 Separation Between Anchor Candidates and Ligand-receptor Models

The final selection illustrates why anchor candidates and ligand-receptor models should be separated. HEPH and BOC passed the top-four single-cell gate but were not retained as final native-ligand model hypotheses because HEPH lacks a clear native extracellular ligand-receptor pair—although iron homeostasis genes have been implicated in IPF [28] and HEPH has documented pathological significance in lung tissue [29]—and BOC carries Hedgehog-associated profibrotic activation risk. In contrast, PLXDC1 and ROBO1 have interpretable extracellular ligand systems that can be evaluated structurally. Even for these two candidates, the current data support hypothesis prioritization rather than target validation.

4.5 Structural Confidence and Hotspot Interpretation

The AF3 models showed modest ipTM scores (0.21–0.36), below the 0.5 threshold for reliable interface prediction. The SLIT2–ROBO1 model was relatively more trustworthy because it was informed by the 2V9T crystal structure. The 17 high-priority hotspot residues provide testable hypotheses: the PLXDC1 cluster (149–156) is suitable for PEDF-binding mutagenesis, and the ROBO1 Ig1 hotspots overlap the validated SLIT2 interface.

4.6 Limitations and Future Validation

This work is computational and hypothesis-generating. Key limitations include dataset differences, HPA annotation gaps, transcript–protein mismatch, and modest AF3 complex confidence (ipTM 0.21–0.36). Predicted structures cannot substitute for experimental binding data. Future work should validate cell-surface expression, ligand binding, receptor internalization and targeted delivery in IPF-relevant models.

5 CONCLUSION

This study prioritized PLXDC1 and ROBO1 as two model-supported membrane-anchor hypotheses through a multi-stage workflow integrating bulk transcriptomics, HPA expression evidence, single-cell refinement and structure-guided analysis. Three model hypotheses (PEDF–PLXDC1, SLIT2–ROBO1, SLIT3–ROBO1) and 17 high-priority hotspot residues were identified for future experimental validation.

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